Contracts

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Name		Domain		Range
`num-sqr`	`::`	`Number`	`->`	`Number`
`num-sqr(9)`				
`num-sqrt`	`::`	`Number`	`->`	`Number`
`num-sqrt(25)`				
`triangle`	`::`	`Number, String, String`	`->`	`Image`
`triangle(80, "solid", "darkgreen")`				
`circle`	`::`	`Number, String, String`	`->`	`Image`
`circle(30, "outline", "fuchsia")`				
`star`	`::`	`Number, String, String`	`->`	`Image`
`star(50, "solid", "teal")`				
`square`	`::`	`Number, String, String`	`->`	`Image`
`square(10, "outline", "red")`				
`rectangle`	`::`	`Number, Number, String, String`	`->`	`Image`
`rectangle(20, 80, "solid", "gold")`				
`text`	`::`	`String, Number, String`	`->`	`Image`
`text("I'm thankful for…", 50, "green")`				
`ellipse`	`::`	`Number, Number, String, String`	`->`	`Image`
`ellipse(30, 70, "outline", "lightblue")`				

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Name		Domain		Range
`# regular-polygon`	`::`	`Number, Number, String, String`	`->`	`Image`
`(regular-polygon 40 8 "solid" "red")`				
`# rhombus`	`::`	`Number, Number, String, String`	`->`	`Image`
`(rhombus 20 50 "solid" "turquoise")`				
`# right-triangle`	`::`	`Number, Number, String, String`	`->`	`Image`
`(right-triangle 20 50 "outline" "black")`				
`# isosceles-triangle`	`::`	`Number, Number, String, String`	`->`	`Image`
`(isosceles-triangle 100 30 "solid" "black")`				
`# radial-star`	`::`	`Number, Number, Number, String, String`	`->`	`Image`
`(radial-star 17 50 10 "solid" "orange")`				
`# star-polygon`	`::`	`Number, Number, Number, String, String`	`->`	`Image`
`(star-polygon 100 10 3 "outline" "yellow")`				
`overlay`	`::`	`Image, Image`	`->`	`Image`
`overlay(star(30, "solid", "gold"),circle(30, "solid", "blue"))`				
`beside`	`::`	`Image, Image`	`->`	`Image`
`beside(star(50, "solid", "orange"),circle(50, "solid", "green"))`				
`above`	`::`	`Image, Image`	`->`	`Image`
`above(triangle(30, "solid", "red"),square(30, "solid", "blue"))`				

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Name		Domain		Range
`put-image`	`::`	`Image, Number, Number, Image`	`->`	`Image`
`put-image(star(30, "solid", "red"), 50, 150, rectangle(300, 200, "outline", "black"))`				
`rotate`	`::`	`Number, Image`	`->`	`Image`
`rotate(35, rectangle(30, 80, "solid", "purple"))`				
`scale`	`::`	`Number, Image`	`->`	`Image`
`scale( 0.8, triangle(30, "solid", "red"))`				
`string-repeat`	`::`	`String, Number`	`->`	`String`
`string-repeat("cheetah ", 5)`				
`string-contains`	`::`	`String, String`	`->`	`Boolean`
`string-contains("rockstar", "star")`				
`num-min`	`::`	`Number, Number`	`->`	`Number`
`num-min(80, 20)`				
`num-max`	`::`	`Number, Number`	`->`	`Number`
`num-max(80, 20)`				
`count`	`::`	`Table, String`	`->`	`Table`
`count(animals-table, "species")`				
`mean`	`::`	`Table, String`	`->`	`Number`
`mean(animals-table, "age")`				

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Name		Domain		Range
`median`	`::`	`Table, String`	`->`	`Number`
`median(animals-table, "age")`				
`modes`	`::`	`Table, String`	`->`	`List<Number>`
`modes(animals-table, "age")`				
`bar-chart`	`::`	`Table, String`	`->`	`Image`
`bar-chart(animals-table, "legs")`				
`pie-chart`	`::`	`Table, String`	`->`	`Image`
`pie-chart(animals-table, "species")`				
`histogram`	`::`	`(t :: Table, column :: String, bin-width :: Number)`	`->`	`Image`
`histogram(animals-table, "age", 2)`				
`box-plot`	`::`	`Table, String`	`->`	`Image`
`box-plot(animals-table, "age")`				
`modified-box-plot`	`::`	`Table, String)`	`->`	`Image`
`modified-box-plot(animals-table, "age")`				
`scatter-plot`	`::`	`(t :: Table, labels :: String, xs :: String, ys :: String)`	`->`	`Image`
`scatter-plot(animals-table, "species", "pounds", "weeks")`				
`image-scatter-plot`	`::`	`(t :: Table, xs :: String, ys :: String, f :: (Row -> Image))`	`->`	`Image`
`image-scatter-plot(animals-table, "pounds", "weeks", animal-img)`				

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Name		Domain		Range
`r-value`	`::`	`(t :: Table, xs :: String, ys :: String)`	`->`	`Number`
`r-value(animals-table,"pounds", "weeks")`				
lr-plot	::	(t :: Table, labels :: String, xs :: String, ys :: String)	->	Image
lr-plot(animals-table, "species", "pounds", "weeks")				
random-rows	::	(t :: Table, num-rows :: Number)	->	Table
random-rows(animals-table, 5)				
<Table>.row-n	::	Number	->	Row
animals-table.row-n(5)				
<Table>.order-by	::	(col :: String, increasing :: Boolean)	->	Table
animals-table.order-by("species", true)				
<Table>.filter	::	(test :: (Row -> Boolean))	->	Table
animals-table.filter(is-cat)				
<Table>.build-column	::	(col :: String, builder :: (Row -> Any))	->	Table
animals-table.build-column("sticker", label)				
bar-chart-summarized	::	(t :: Table, labels :: String, values :: String)	->	Image
bar-chart-summarized(animals-table, "species", "pounds")				
pie-chart-summarized	::	(t :: Table, labels :: String, values :: String)	->	Image
pie-chart-summarized(animals-table, "age", "pounds")				

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Name		Domain		Range
`grouped-bar-chart`	`::`	`(t :: Table, group :: String, subgroup :: String)`	`->`	`Image`
grouped-bar-chart(animals-table, "sex", "species")				
stacked-bar-chart	::	(t :: Table, group :: String, subgroup :: String)	->	Image
grouped-bar-chart(animals-table, "sex", "species")				
	::		->	
	::		->	
	::		->	
	::		->	
	::		->	
	::		->	
	::		->	